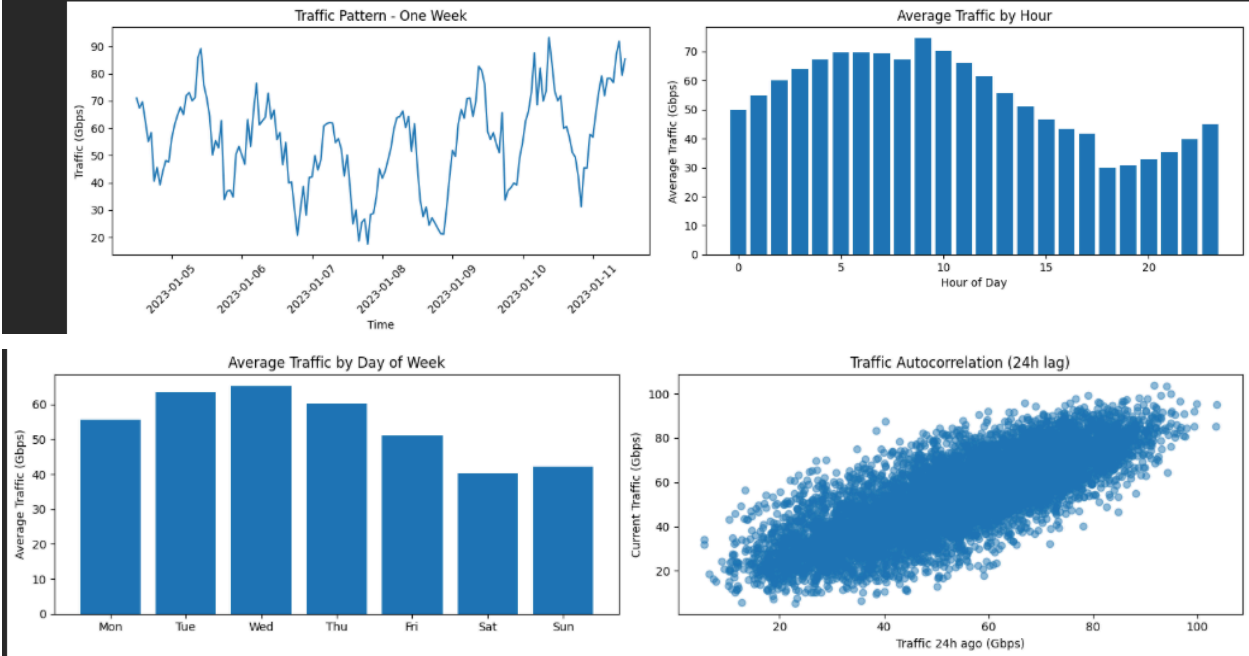


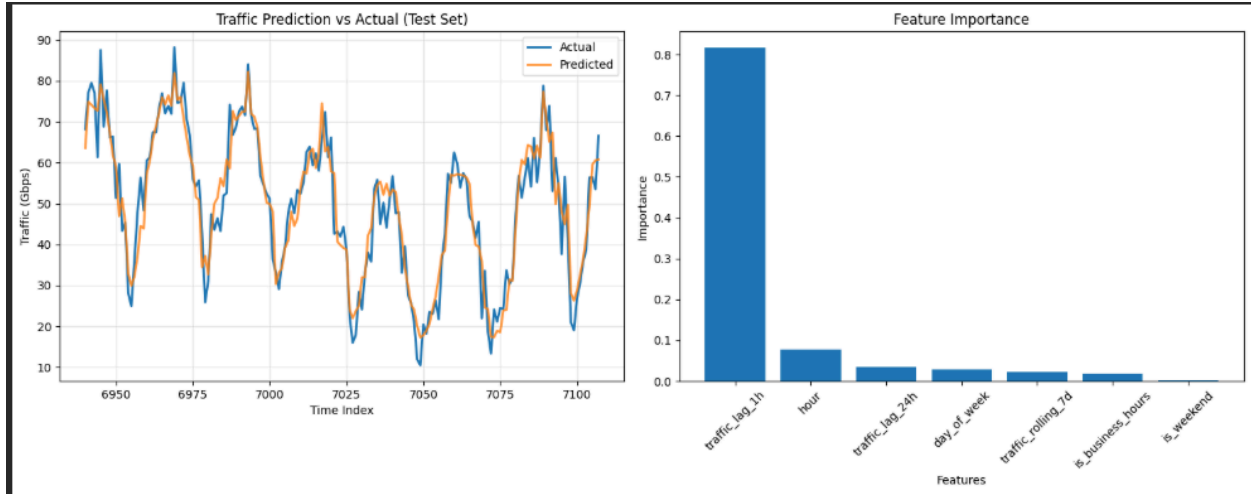
```
First few rows:
...
timestamp hour day_of_week is_weekend is_business_hours \
84 2023-01-04 12:00:00 12 2 False True
85 2023-01-04 13:00:00 13 2 False True
86 2023-01-04 14:00:00 14 2 False True
87 2023-01-04 15:00:00 15 2 False True
88 2023-01-04 16:00:00 16 2 False True

traffic_gbps traffic_lag_1h traffic_lag_24h traffic_rolling_7d
84 71.050823 77.678321 70.680542 54.166101
85 67.408125 71.050823 66.971737 54.147588
86 69.670302 67.408125 57.544738 54.132844
87 62.594911 69.670302 52.953244 54.090667
88 55.123982 62.594911 59.818534 54.024632
```



```
Model Performance:
Training MSE: 17.59
Testing MSE: 34.64
Training R²: 0.946
Testing R²: 0.893

Feature Importance:
feature importance
4 traffic_lag_1h 0.815706
0 hour 0.077702
5 traffic_lag_24h 0.035665
1 day_of_week 0.027819
6 traffic_rolling_7d 0.023439
3 is_business_hours 0.017457
2 is_weekend 0.002211
```



A dataset was generated to simulate hourly telecom network traffic over one year.

The dataset contained hourly network traffic data with daily, weekly, and seasonal patterns. Features I found that were used in this dataset is, hour of day, day of week, etc.

- traffic_lag_1h represents network traffic for one hour
- traffic_lag_24h represents the same hour on the previous day
- traffic_rolling_7d is to calculate the traffic over seven days

Higher traffic occurs during business hours while during nighttime has lower traffic

The machine model used to create the dataset is RandomForestRegressor

Results show the model did predict network traffic learning about the daily cycles, weekly cycles and the traffic behavior. The prediction graph shows that predicted traffic closely follows actual traffic patterns. This shows that the model successfully captured the network usage.